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PREFACE

Participatory Design meets Artificial Intelligence.

Co-imagining mutual learning of AI technologies and designing with AI tools.

1. Introduction

This Special Issue (N. 68) emerged as a step toward broadening the discussion about the relationship between Participatory Design (PD) and Artificial Intelligence (AI). The original PD projects were motivated by the introduction of computer technologies into industrial workplaces, where they threatened to impoverish or take over jobs. Just like computers were a threat to work and workers in the 1970s and 80s, AI is seen as a threat to workers—and societies—today. Building on a workshop held at NordiCHI 2024 [1], we wanted to invite a wider community of researchers and practitioners to critically reflect on how PD can question, shape, and reimagine AI technologies and practices.

The notion of AI was used for the first time in 1955 referring to a machine that can simulate human intelligence [2]. Webster [3] argues that current AI systems that are based on Machine Learning (ML) and Large Language Models (LLMs) can "be said to fulfill the original 1955 definition of AI for the first time" (p. 8). Today, AI systems are widespread and deployed rapidly across domains of considerable social significance—in healthcare, education, employment, criminal justice, and many others—without appropriate safeguards or accountability structures in place [4,5]. AI can replace human performance of tasks and "cooperate" with humans [6]. Similar to the expert systems of the second wave of AI [3], the current third wave of AI advocates AI that improves human capabilities rather than replacing them. The initiative for Human-Centered AI (HCAI) argues that AI systems should be explainable, comprehensible, useful and usable [7]. To achieve this Shneiderman [8] argues that user-centered and participatory methods engaging stakeholders in design is needed. A study of research on HCAI by Capel and Brereton [9] found that almost half the research papers could be categorized as human-centered approaches to design of AI, but very few of these were based on PD [10]. This Special Issue aims to change this.

PD is an approach to design that aims to involve future users and stakeholders in the whole span of the design process [e.g. 11]. Participation in PD means enabling users to make design decisions [12] through processes of mutual learning and co-creation [13]. An important aspect of PD is to ground the design process in the use context to secure a solution that supports the users in their activities [14,15]. PD has an explicit political ambition by aiming to enhance users' autonomy and power of their environment [10, 11,12,16].

The seven selected papers in this Special Issue broadly reflect on co-design of future ethical, responsible, useful and supportive AI solutions. They present a range of perspectives and visions on how participatory approaches can be used to design AI and how AI can be used to support PD. The papers in the Special Issue combine AI and PD in these two ways: (1) using PD as an approach to co-design AI systems allowing users to influence their future digital environment, and (2) using AI tools in PD supporting users and designers in collaborative design co-creating and co-designing the digital system.

PD for AI: PD as an approach to co-design AI systems. Based on the original PD's democratic ambition to strengthen workers' control of their work environment and work tools, we wanted to explore if PD could be used as approach for designing AI-infused digital solutions that support the users' activities and enhance their autonomy and control of their (work) environment [17,18]. PD's emphasis on involving users in design as experts on their own practices can also be practiced when the digital solution includes AI [15,19]. Bratteteig and Verne [20] argue that PD is well suited for discussions between designers and users negotiating and mitigating the challenges AI poses to users' practices. The socio-technical complexity often associated with novel AI systems requires a rethinking of how we ensure end-users remain voiced in important decision-making [21]. The framing of participation within the fast stream of new AI discourse also tends to be less concerned with some of the original ideas of PD, which has raised concerns about e.g., "participation washing" [22] or how PD can challenge AI solutionist approaches [14]. We argue that there is still a need for PD approaches aimed at understanding the technology and its potential for changing workplaces and work practices, as well as to open up for people to have a say in choices concerning the technology during its design and use [23].

AI for PD: AI tools in PD supporting collaborative design. In general, PD research has focused on developing tools and methods for user participation in design, including support for facilitating mutual learning, co-design and user involvement in design decisions. There is a growing body of research exploring how AI tools can enhance and support users (and designers) in PD enhancing the users' influence and engagement in design. In particular, design practices involving human-AI collaboration have sparked a stream of new research, resurfacing tensions concerning decision-making when AI is involved in the design process (see, e.g., [24,25]). This includes fear of reducing users' ability to control and shape the design space explored in design processes, e.g., by introducing design fixation [26], reducing the human role to that of an evaluator of AI output [27], or directly limiting how choices are made [24]. As such, PD plays an important role in reminding us of the purpose of involving end-users, how we should address some of the growing concerns [28], and how these values are being challenged by new ways of conceptualizing participation [22].

2. Papers in the special issue.

The papers in this Special Issue address the two ways of combining PD and AI discussed above. To our surprise, none of the papers were grounded in the "classic" Scandinavian PD tradition signalling both that PD means different things to different designers and that PD methods have spread and are considered useful in many nuances of design. We recognize PD's emphasis on understanding users' practices as a basis for designing suitable technical support, and how designers work with translating values and ethical principles based in the use context into the AI's data and algorithms. The first four papers elaborate on how PD can be used for designing better AI. The last three papers focus on the PD process itself and how AI can support and influence the participation as well as the design process. The sequence of papers reflects how strong the PD perspective is represented in each of the papers.

2.1 PD for AI

The paper "PD-with/for-AI: Framework and Lessons for Responsible Use of AI-Generated Synthetic Personas" [29] presents a conceptual model for PD-with/for-AI, where Participatory Design (PD) and generative AI (GenAI) have a coupled and bidirectional relationship enabled by AI-generated and enacted synthetic personas. The authors argue that PD is well-suited to evaluate AI systems through lived experiences, surfacing bias across symbolic, structural, and individual layers, and that these insights can motivate the careful introduction of GenAI-supported co-creation tools within participatory processes. Findings are drawn by the synthesis of four previous studies showing how synthetic personas function best as situated provocations and thinking partners, capable of widening perspectives, unsettling assumptions, and supporting inclusive design ideation. However, their use is constrained by key risks, including bias, stereotyping, and the believability hazard, where fluent and convincing outputs may be mistaken for authoritative or grounded knowledge. The paper formalizes a non-substitution principle, emphasizing that synthetic personas are adjuncts that open dialogue and structure inquiry, but must not replace engagement with affected stakeholders or be used to justify design decisions. The authors outline a set of care practices as a way to enhance responsible use, including co-creation and/or community validation, explicit disclosure and consent, provenance and documentation, and reflexive facilitation. These elements are integrated into a layered, bidirectional model, where PD values such as inclusion, equity, and accountability impose normative constraints on care practices and instruments, while insights from the use of GenAI tools feed back into the ongoing reconfiguration of PD practices and values. Overall, the paper positions PD as both shaper and steward of GenAI.

The paper “Adapting Experience-Based Co-Design to Inform AI Support Tools: Nurses’ Perspectives on Handover” [30] describes a PD project in which the authors expand the healthcare-oriented service design method Experience-Based Co-Design (EBCD) with methods for designing AI support for and with nurses. The authors provide a detailed description of how the PD process was carried out providing detailed descriptions of the mutual learning and the codesign activities. They demonstrate how they maintained having nursing practice in the foreground throughout the process. The paper also shows how the collaboration between nurses, designers and AI experts resulted in design of AI support for nursing handovers that met nurses’ needs and was technically feasible. The paper shows how service design and PD overlap when taking people’s practices and activities as the starting point for identifying problems and their possible solutions. The AI design workshops included in the EBCD method add to PD research by providing an instructive example of design of AI support for a complex practice carried out as a multidisciplinary collaboration between technologists, designers and future users.

The paper “Co-Designing AI-Enhanced E-Participation. Insights from ORBIS Participatory Approaches to Democratic Deliberation” [31] analyses how PD operates as a socio-technical mediation mechanism in the development of AI-enhanced e-participation platforms for democratic deliberation. Drawing on the Horizon Europe ORBIS project, which embedded PD practice throughout the entire project lifecycle, the authors follow six pilots to offer an empirical account of mutual learning and co-design with AI in the public domain. The analysis traces how differentiated aims and practices translate into specific AI design choices, and how this process gradually reconfigures roles and workflows among project stakeholders and civic representatives. The paper proposes a theoretical account structured around four intersections (goals × deliberative purposes, scope × AI configuration, participants × role reconfiguration, and methods × iterative evolution) that constitute analytic dimensions for unpacking how PD unfolds in AI-enhanced deliberation as longitudinal mediation rather than a set of discrete events. The work foregrounds how AI-enhanced activities actively shape what becomes visible and actionable for participants in civic processes, and the paper reflects on how PD can help redistribute interpretive authority, surface frictions, and address institutional misalignments and uptake.

The paper “Suitability of Responsible AI Assessment Tools for MSMEs” [32] presents a participatory study of the suitability of Responsible AI (RAI) assessment tools for Micro, Small, and Medium Enterprises (MSMEs), focusing on whether existing tools effectively support their practical needs, constraints, and levels of AI maturity. The study is based on a structured analysis of a diverse set of tools originating from government bodies, academia, private sector actors, and international organizations. It evaluates these tools against criteria derived from literature on trustworthy AI and AI governance, such as usability, provision of practical guidance, alignment with regulatory frameworks like the EU AI Act and GDPR, and their ability

to translate high-level ethical principles into actionable steps. The findings reveal that while many tools emphasize risk management and regulatory compliance, they often lack accessibility, pedagogical support, and contextual adaptability required by MSMEs, particularly in terms of simplifying complex requirements, reducing long textual content, and enabling step-by-step implementation. Furthermore, most tools do not sufficiently encourage ecosystem-building, collaboration with academia, or mechanisms for explaining AI outcomes, which are critical for fostering responsible AI adoption in smaller organizations. Hence remaining a significant gap in providing lightweight, user-friendly, and implementation-oriented solutions tailored to MSMEs. The authors conclude that current RAI assessment tools are only partially suitable and highlights the need for more inclusive, scalable, and practice-oriented approaches that bridge the gap between high-level ethical frameworks and the operational realities of MSMEs, thereby supporting more effective and responsible AI governance.

2.2 AI for PD

The paper “AI, Metaverse and co-design for social innovation: an application protocol” [33] explores how the integration of generative AI and metaverse technologies can extend and reshape co-design practices within social innovation processes (drawing on the OSMOSI project conducted in Milan and Palermo). The researchers develop and test a replicable application protocol that combines AI-driven data analysis, generative visualisation, conversational interfaces, and immersive virtual environments to support mapping, co-design, and dissemination phases. The project shows how AI can act as an interpretative, generative, and mediating tool that enhances knowledge synthesis, accessibility, and toolkit production, while the metaverse creates inclusive, persistent spaces for distributed and asynchronous participation. At the same time, the paper critically addresses issues of bias, accessibility, and the reconfiguration of roles in participatory processes. The contribution fits the special issue by offering both empirical evidence and a structured methodological framework for integrating AI into co-design, advancing understanding of how emerging technologies can augment collaboration, inclusivity, and governance in complex social innovation contexts.

The paper “Imagining climate-resilient urban spaces: Integrating AI into community consultation in Samarinda” [34] examines how generative artificial intelligence can be integrated into participatory design processes to support more inclusive and iterative community engagement, using a case from urban co-design workshops in Samarinda, Indonesia. By embedding a text-to-image AI tool (Midjourney) within a structured participatory design workflow, the project demonstrates how AI can function as both an “imagination engine” and a boundary object that enables participants to co-create, visualize, and refine climate-adaptive public space proposals in real time. The findings highlight how AI-supported visualisation can broaden participation across age groups and language barriers,

accelerate feedback loops, and make participants' influence more visible in the design process, while also raising questions around facilitation, bias, and responsible use. The paper fits the special issue by critically exploring how AI can be utilised in participatory design workshops, offering both empirical insights and practical guidelines for embedding generative AI in co-design practices, and contributing to ongoing discussions about how AI can reshape participation, creativity, and decision-making in design processes.

The paper "Human–AI Collaboration in UX: Rethinking Creativity and Productivity in Understanding User Needs" [35] investigates how UX professionals adopt and perceive AI tools in their creative workflow. Their research examines the role of AI in practice, particularly during discovery activities, and how its use influences creativity and productivity, highlighting both benefits and risks, such as overreliance and loss of originality. It further seeks to explore human-AI collaboration in UX by investigating its impact on individual practitioners and teams through surveys, interviews, and comparative experiments. While they show that designers widely adopt AI to shift their focus from repetitive tasks to more strategic activities, their findings also complicate these perceptions: the use of AI in co-creative workflows may lead to a "productivity illusion," where designers feel more productive without consistently producing better outcomes. It also problematizes how the use of AI can potentially distance designers from the emotional aspect of understanding users. Their empirical findings offer valuable insight that can inform PD debates by surfacing how practitioners negotiate AI's promises of efficiency with concerns about mutual learning, e.g., accountability, transparency, and empathy. Their work reflects on the reconfigurations in designers' workflows that further address relevant PD concerns such as the tradeoff between efficiency and deep engagement, new forms of dependency or overreliance on AI tools, and differentiated power dynamics.

3. Concluding remarks

This Special Issue offers a diverse set of perspectives on how PD can both shape AI and be shaped by AI. The contributions show that PD is not an optional complement to AI, but a necessary approach for grounding AI in situated practices, enabling mutual learning, and strengthening democratic control over technological change. Across various domains—from healthcare to social innovation, climate resilience, and UX practice—the papers illustrate how PD can support responsible, useful, and accountable AI, while also revealing tensions such as design fixation, overreliance on AI tools, and shifting power dynamics in design processes. Each article sheds light on different ways that AI and PD intersect: using PD to co-design AI systems; using AI to support and expand participation; and critically probing how AI alters roles, workflows, and understandings of creativity and responsibility.

For the PD research community this Special Issue contributes to current PD research by addressing technical development as part of and result of PD: the papers give examples of computational alternatives [36]. Today, computational systems and alternatives are very different from the early PD days (see e.g., [37,38]). However, there is still much to learn from the early PD projects, in particular their emphasis on "local action for a global strategy" [39,40]. This Special Issue is one step in a strategy towards technical changes that we want [41-44]. The papers can serve as scholarly references, but also as inspiration for researchers, practitioners, and policymakers who seek to develop more inclusive and democratic forms of human–AI interaction. It is important to keep PD's democratic ambitions alive in the rapidly evolving AI landscape.

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